

WID3002 NATURAL LANGUAGE PROCESSING, SEMESTER 2, 2025/2026

TOXIC COMMENT DETECTOR

for English, Malay and Manglish

Group 8:

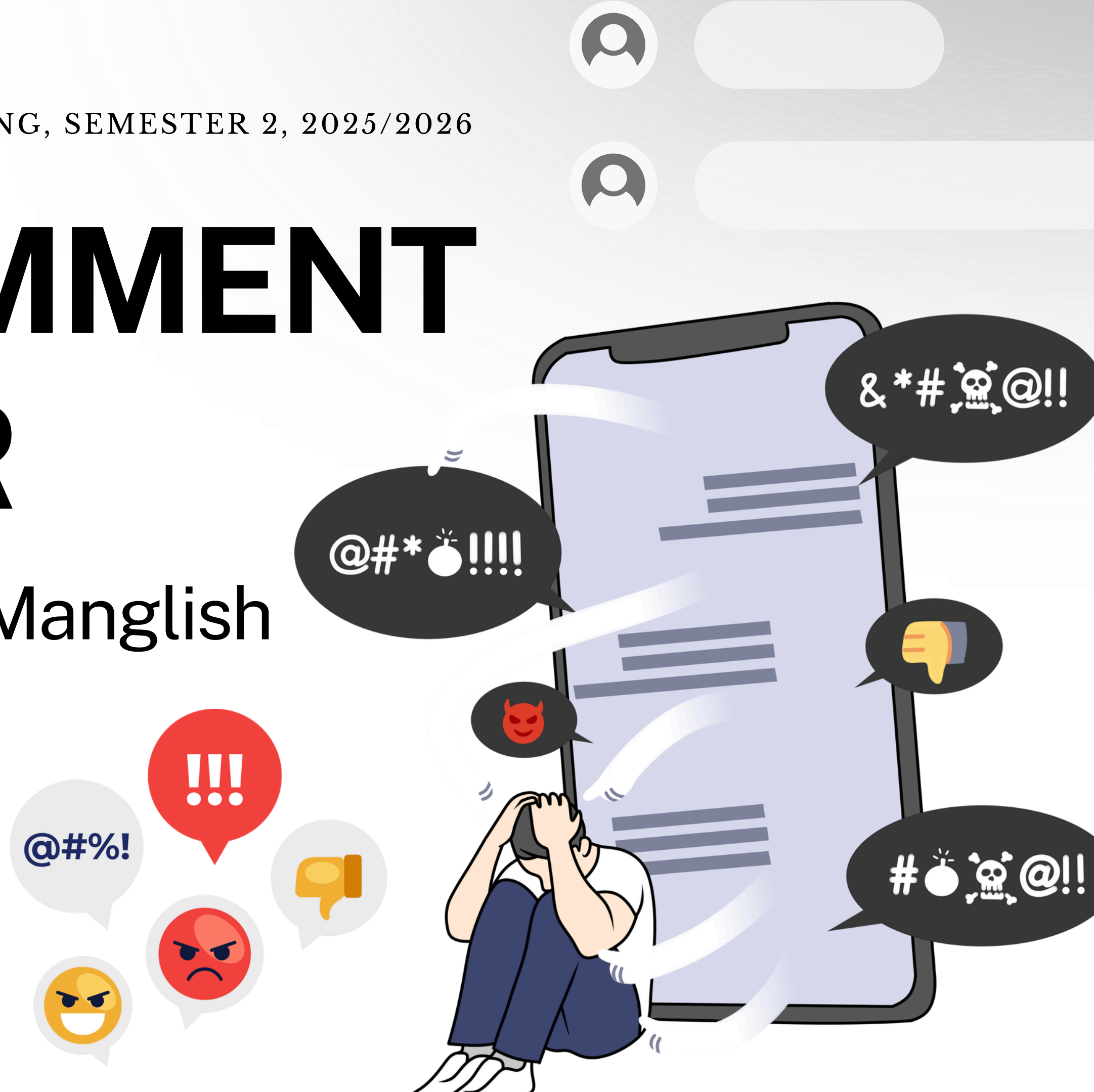
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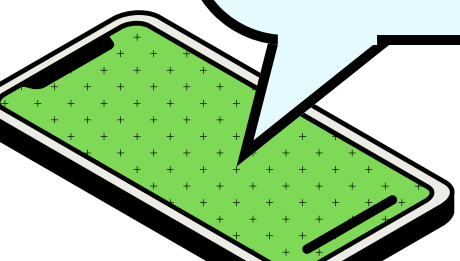


Introduction

Rapid expansion of social media has made user-generated content a communication staple, yet platforms are increasingly overwhelmed by toxic language that poisons digital interactions.

Research shows that nearly **13% of analyzed comments are harmful**, and manual moderation is no longer scalable.

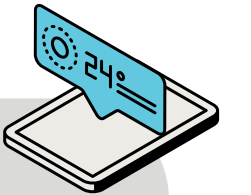
Our project addresses this by developing an intelligent classifier using **XLM-RoBERTa**, designed to navigate the linguistic complexities of English, Malay, and code-mixed **Manglish** to foster a more respectful online communities.



Problem Statement

Moderation Crisis

Over 25,000 cases of harmful content (MCMC, 2023) make manual moderation impossible.



Linguistic Barrier

Traditional NLP (BOW/Word2Vec) fails to capture the semantic nuances of local Malay slang and English abbreviations.



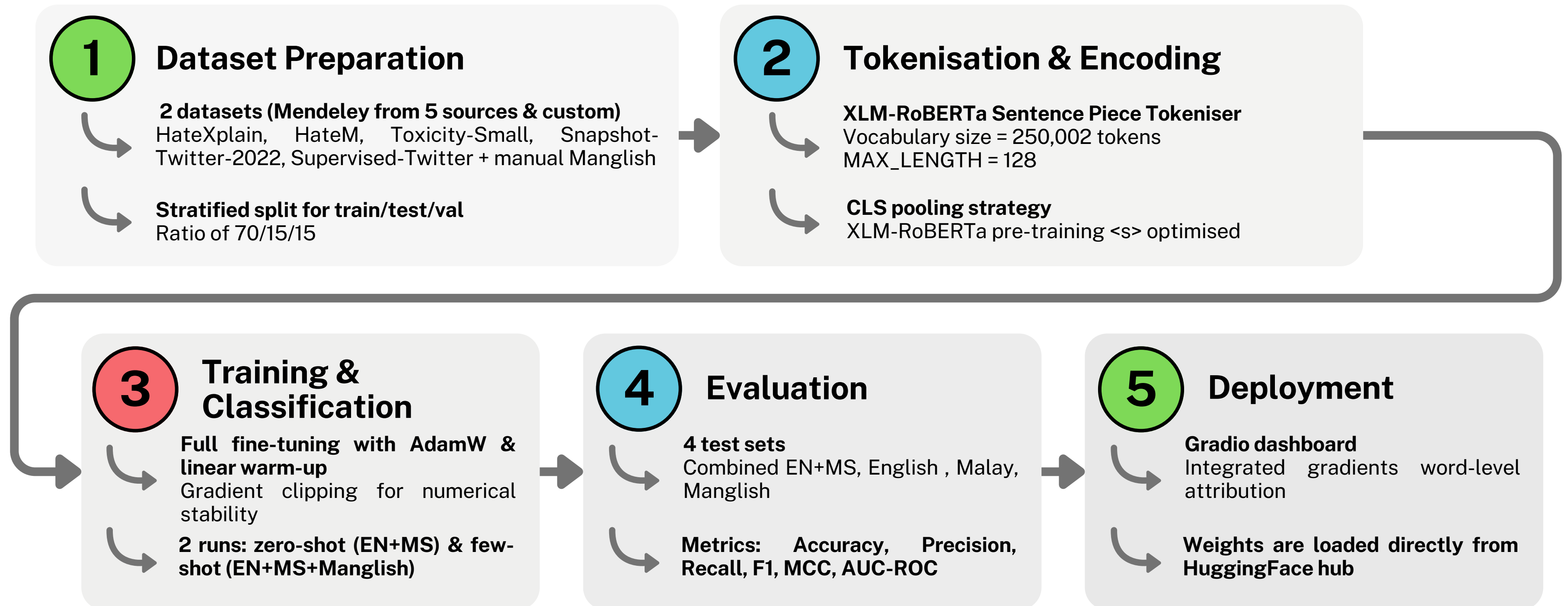
The Manglish Gap

Standard transformer models lose contextual accuracy when switching between English and Malay within a single sentence.



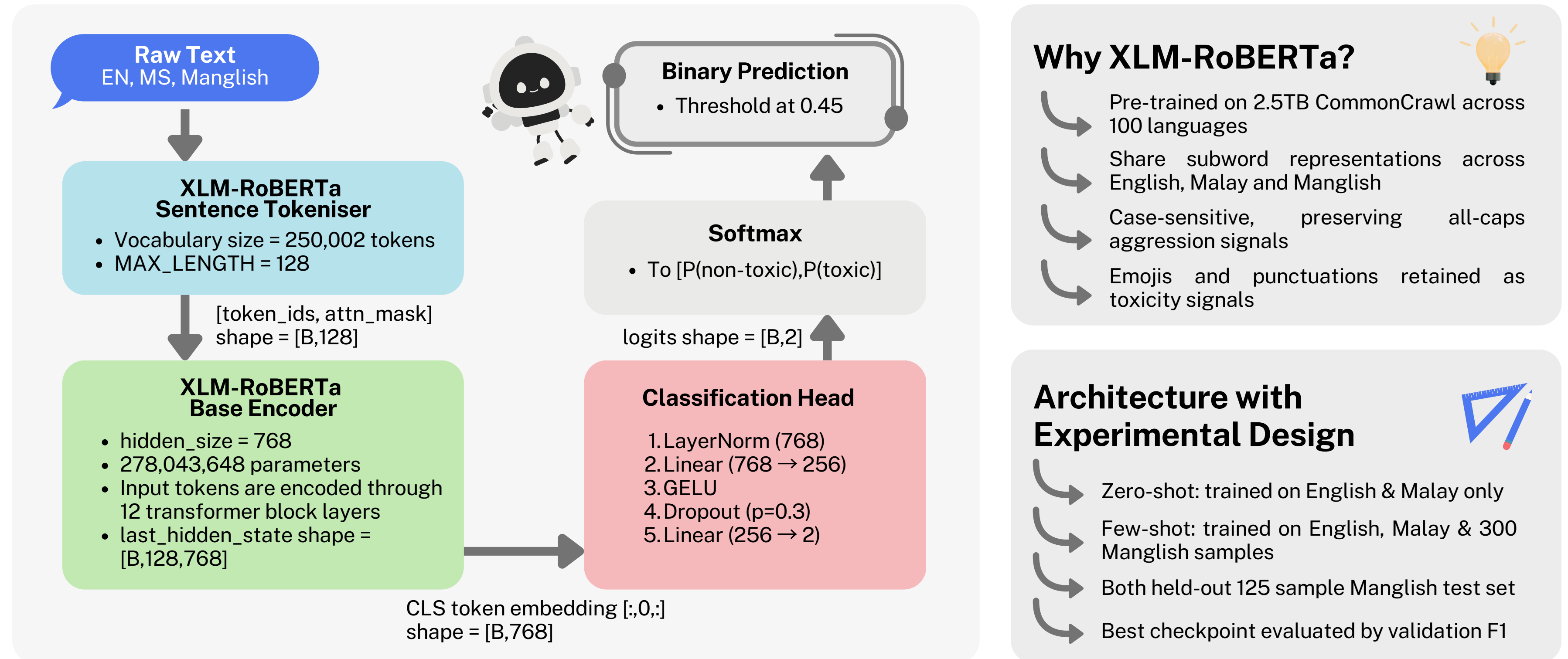
Solution Design & Pipeline

The below shows the five-stage pipeline of our toxic comment detector, centred around XLM-RoBERTa Base.

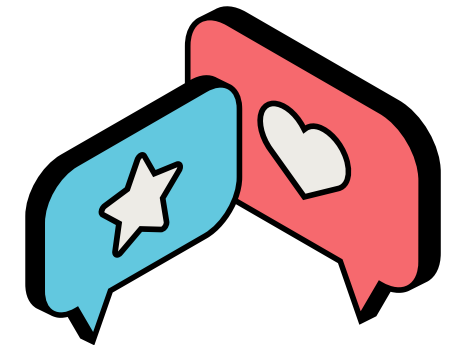


Solution Architecture

The architecture shows the internal structure of the model from raw text input to binary prediction.

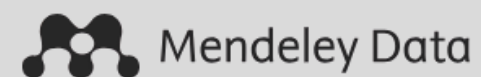


Dataset & Preprocessing



Data Sources

Primary Dataset:



Bilingual Malay-English Social Media Dataset for Binary Hate Speech Detection

- 26,985 bilingual english and malay social media posts (5 sources)
- 14,642 non-hate and 12,343 hate-labeled entries
- 13,609 English and 13,376 Malay texts
- Used for training the base model

Custom Dataset (Team-curated):



Custom Code-mixed Malay-English (Manglish) Dataset

- 500 code-mixed sentences from Reddit Malaysia, X, Tiktok and Facebook
- 274 toxic and 226 non-toxic labeled entries
- Used for few-shot fine-tuning on Manglish
- Makes the model relevant to Malaysian usage

Preprocessing

Text Cleaning:

- What we normalize:
 - Smart quotes
 - Emoticon placeholder tags
- What we keep:
 - Casing
 - Emoji
 - Special characters and symbols (Punctuations, Apostrophes)
 - User-typed bracket content
- What we removed:
 - URLs
 - @mentions
 - Anonymization placeholder tag
 - newline characters
 - Repeated & leading whitespace

Splitting & Experimental Setup:

Stratified 70 / 15 / 15 split (by language + label)

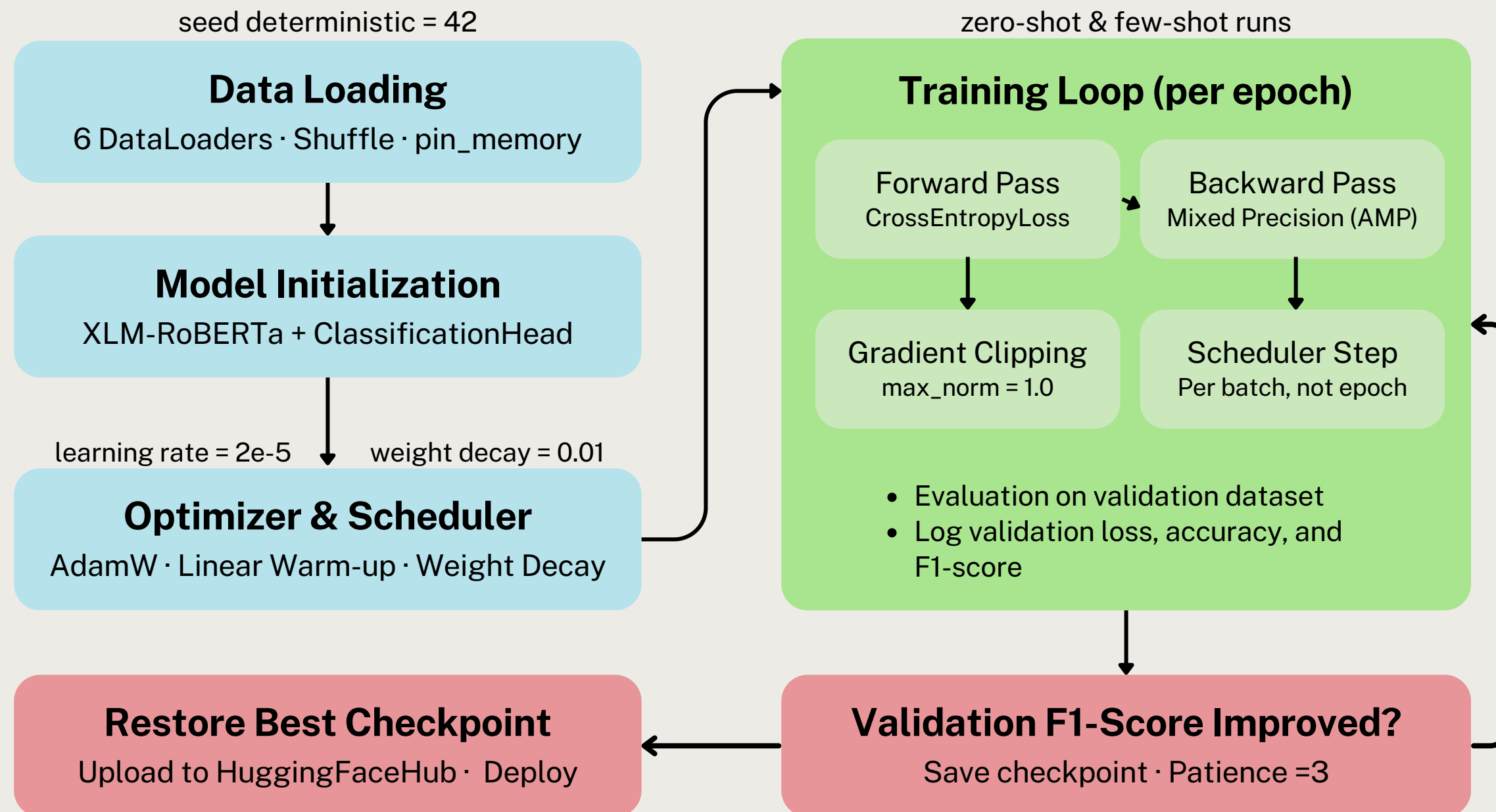
- Zero-shot (baseline)
 - Train EN + MS only
 - No Manglish in training
- Few-shot (fine-tuned)
 - Train EN + MS + 300 Manglish
 - +75 Manglish validation

Same 125 Manglish test set for both

- to compare the performance on manglish text

Training Procedures & Tools

Training Procedures



Tools

Python



Implementation
Language

Google Colab



GPU Training
Environment

PyTorch



Deep Learning
Framework

HuggingFace
Transformers



Pretrained Model &
Tokeniser

Scikit-learn



Evaluation Metrics

Gradio

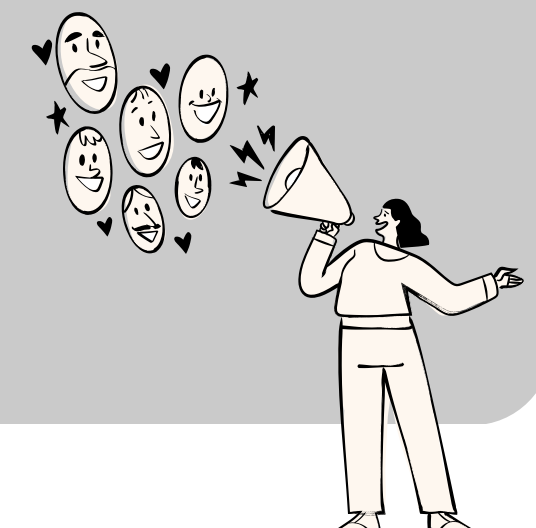


Frontend
Dashboard

GitHub



Dataset Hosting



Results: Per-Language Performance



RESULTS

Per Language Performance

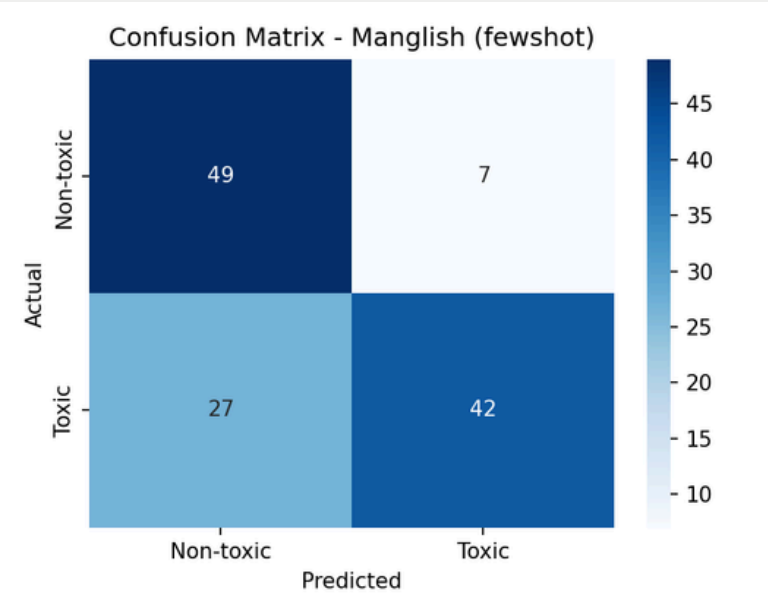
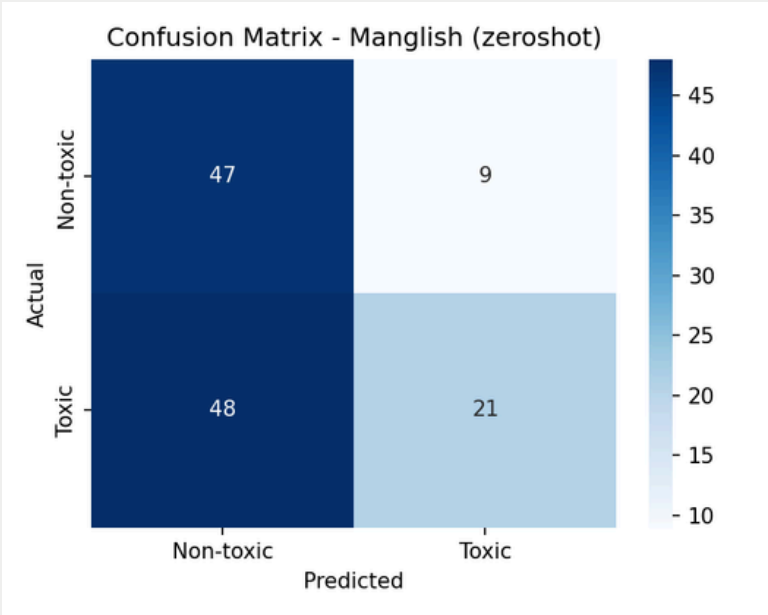
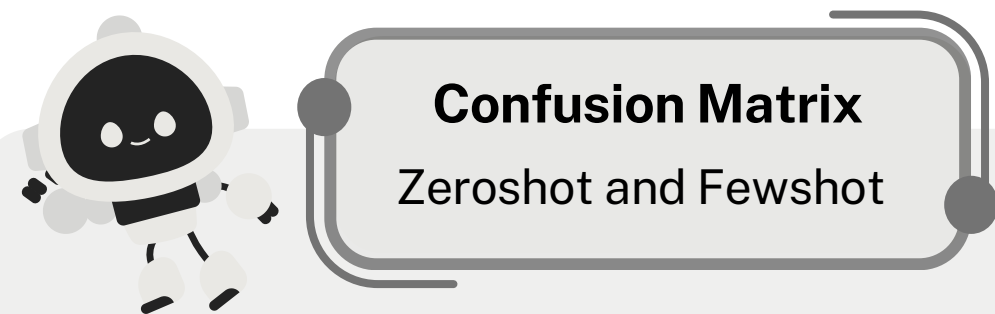
Dataset	Accuracy	Precision	Recall	F1	MCC	AUC-ROC
Combined Test	0.8488	0.8202	0.8575	0.8384	0.6971	0.9258
English Test	0.8487	0.831	0.8337	0.8323	0.6945	0.9172
Malay Test	0.849	0.8105	0.8809	0.8442	0.7004	0.9311
Manglish Code-Mixed Test	0.728	0.8571	0.6087	0.7119	0.4927	0.808

KEY FINDINGS

- Malay achieved the highest overall performance
- Combined F1-score = 83.8%
- Manglish remained the most challenging dataset
- Strong multilingual generalisation across languages



Key Result: Confusion Matrix & Effect of Code-Mixed Fine-Tuning



Model	Accuracy	Precision	Recall	F1	MCC	AUC-ROC
Zero-shot	0.544	0.7	0.3043	0.4242	0.1672	0.7089
Few-shot	0.728	0.8571	0.6087	0.7119	0.4927	0.808

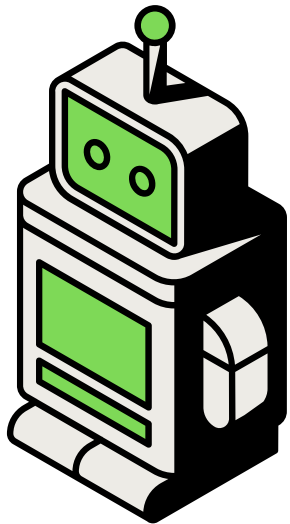
Key Findings from Confusion matrix

- Toxic comments correctly detected increased from 21 → 42
- Missed toxic comments reduced from 48 → 27
- False positives remained low (9 → 7)
- Few-shot model shows stronger detection of toxic Manglish comments

Key Findings from the Table

- F1-score improved from 0.424 → 0.712 (+68%)
- Recall doubled from 0.304 → 0.609
- MCC improved from 0.167 → 0.493
- Only 300 Manglish samples were added for fine-tuning

Live Demo



Multilingual Toxic Comment Detector Dashboard

WID3011 Natural Language Processing System Deployment Sandbox

[Diagnostic Core](#) | [Evaluation Metrics Viewport](#)

Text Input Stream

Type English, Malay, or code-mixed Manglish commentary...

Sensitivity Threshold

0.1

0.45

0.9

Reset Form

Run AI Diagnostics

Verdict

Certainty Score

Distribution

Word-Level Toxicity Attribution

System Processing Telemetry Status Updates

Language-Specific Quick-Test Examples

us English	my Malay	my Manglish
EN Clean	MS Clean	MX Clean
EN Toxic	MS Toxic	MX Toxic

Repository:



Live on
Hugging Face:



Conclusion & Future Improvements

Conclusion

- Strong multilingual performance at **~85% accuracy** across both English and Malay
- Malay achieved the best results with **F1 of 0.844** and **AUC-ROC of 0.931**
- Few-shot fine-tuning with only 300 samples boosted Manglish F1 from **0.42 → 0.71**
- Manglish **remains challenging** due to code-mixing, sarcasm, and limited training data

Future Improvements

- **Expand** Manglish training data through crowdsourcing and augmentation
- **Add language tags** [EN] [MS] [CM] to reduce cross-language contamination
- Incorporate **irony and in-group speech annotations** to fix English false positives
- Extend to **multi-class detection** — hate speech, threats, racism, personal attacks
- Explore **XLM-RoBERTa-large** and continual learning on live platform traffic

